34 miles (55 km)—**the average width of the 1,850-mile (3,000-km) long East African Rift System** of active faults.

Tectonic Earth

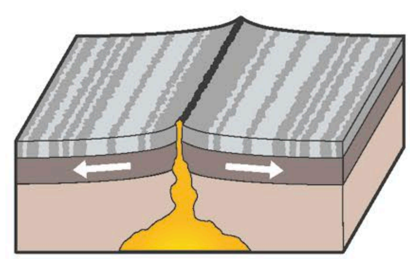
Earth's surface is a layer of solid rock split into huge slabs called tectonic plates, which slowly shift, altering landscapes and causing earthquakes and volcanoes.

The tectonic plates are made up of Earth's brittle crust fused to the top layer of the underlying mantle, forming a shell-like elastic structure called the lithosphere. Plate movement is driven by convection currents in the lower, viscous layers of the mantle—known as the asthenosphere—when hot, molten rock rises to the surface and cooler, more solid rock sinks. Most tectonic activity happens near the edges of plates, as they move apart from, toward, or past each other.

Continental drift

Over millions of years, continents carried by different plates have collided to make mountains, combined to form supercontinents, or split up in a process called rifting. South America's east coast and Africa's west coast fit like pieces of a jigsaw puzzle. Similar rock and life forms suggest that the two continents were once a supercontinent.

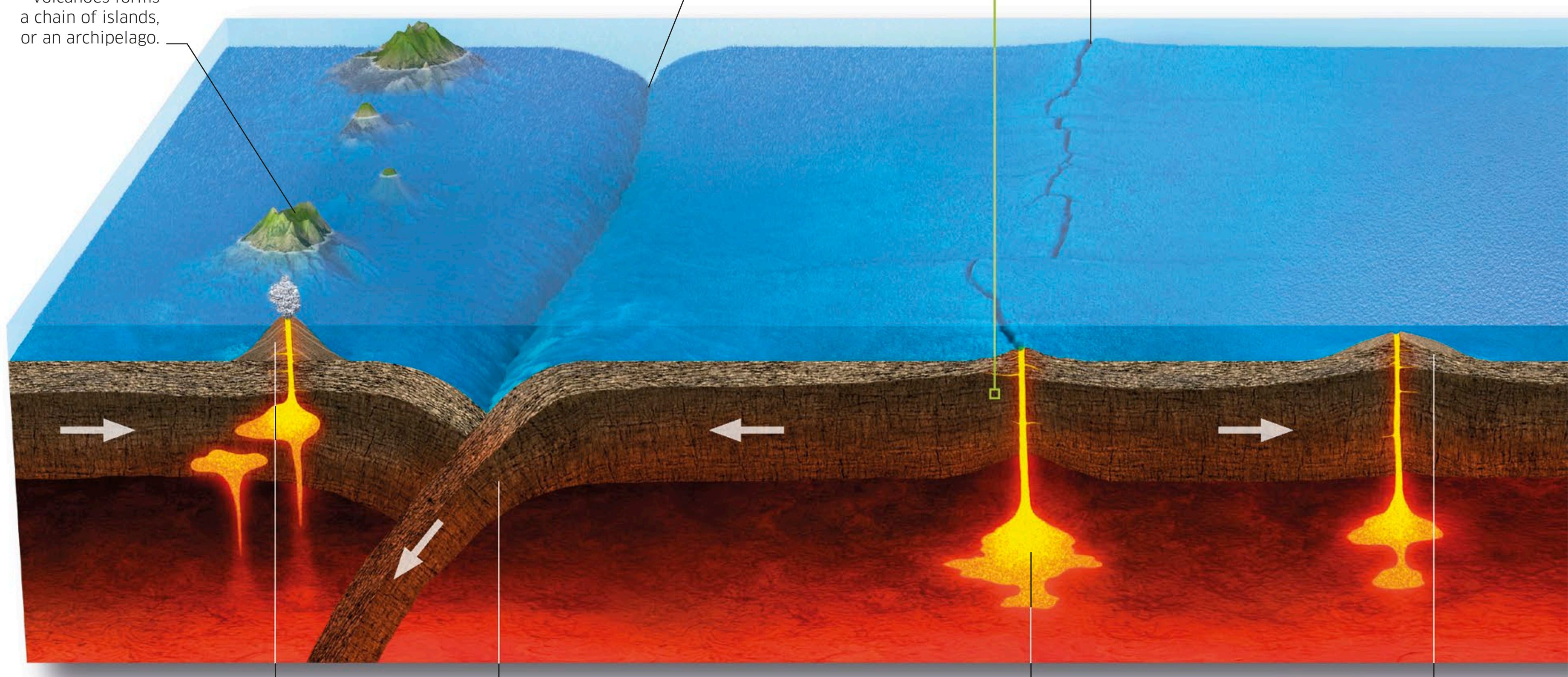
Plates move at between ¼ in (7 mm) per year, one-fifth **the rate human fingernails grow,** and 6 in (150 mm) per year—the rate human hair grows.



Divergent boundary
As two plates move apart, magma welling up from the mantle fills the gap and creates new plate. Linked with volcanic activity, divergent boundaries form mid-ocean spreading ridges under the sea.

Plate tectonics

Where plates meet, landscape-changing events, such as island formation, rifting (separation), mountain-building, volcanic activity, and earthquakes take place. Plate boundaries fall into three main classes: divergent, convergent, and transform.



Island arc
A series of underwater volcanoes forms a chain of islands, or an archipelago.

Ocean trench
Two ocean plates subduct to form a deep-sea trench.

Mid-ocean ridge
Magma wells up as plates move apart, forming a ridge on the ocean floor.

Strato volcano
Layers of hardened lava and ash build up, making these volcanoes steeper than shield volcanoes.

Ocean-ocean subduction
At a convergent boundary under the sea, one oceanic plate slides under the other, creating a mid-ocean trench.

Hot spot
Heat concentrated in some areas of the mantle can erupt as molten magma.

Shield volcano
A shield volcano is built almost entirely of very fluid lava flows, making it quite shallow in shape.